HRISHIKESH BELAGALI

Undergraduate Student

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SUMMARY

Undergraduate computer science major with research interests in quantum algorithms, molecular dynamics, reservoir computing, quantum dynamics, and high performance computing.

SKILLS

Languages: Python, C, C++, LaTeX

Packages: NumPy, PyTorch, scipy, Qiskit, Cirq, PyEMMA, hyperopt, VMD-Python

Softwares: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S) — **Michigan State University Honors College**
Expected Graduation: 2027
Current GPA: 4.0

EXPERIENCE

2025– **Honors Research Scholar** — **MSU Honors College**
- Conducted research on generalized quantum simulators under the guidance of Professor Ryan LaRose.
- Investigated computational group theory methods for subgroup membership in unitary subgroups.

2024– **Undergraduate Research Assistant** — **Vermaas Lab, Plant Research Laboratory**
- Investigated the interaction of the chloroplast targeting protein (At4g16060) with thylakoid membranes using molecular dynamics.
- Analyzed the effect of PDLP-5 (Plasmodesmata Located Protein) on water flow rates through PIP1;4 aquaporin tetramer using grid-steered molecular dynamics.
- Simulated the effects of small molecules like butanol on *Clostridium acetobutylicum* membrane dynamics.

CONFERENCES

2025 *PRL, MSU* — **Aquaporin Water Transit Modulation via Protein Capping** — **acs.digitellinc.com**
Abstract presented at ACS Fall 2025 Conference

2025 *PRL, MSU* — **Evaluating the Interaction of Chloroplast-targeting Protein (At4g16060) with Thylakoid Membrane**
Abstract presented at First Great Lakes Plant Science Conference

PROJECTS

Python, Qiskit — **Discretized Quantum Adiabatic Evolution** — **github.com**
Implemented a quantum adiabatic algorithm to solve QUBO type problems using the Trotter-Suzuki approximation and quantum Ising model formulation techniques

Python, Qiskit — **Quantum Arithmetic Circuits** — **github.com**
Built quantum circuits to perform addition, subtraction and multiplication using the Quantum Fourier Transform.

Python, NumPy — **Monte Carlo Integration** — **github.com**
Monte Carlo integration with importance sampling to numerically evaluate complex integrals.

Python, NumPy — **Langevin Monte Carlo** — **github.com**
Metropolis-adjusted Langevin algorithm for sampling from intractable probability distributions.

Python, OVITO, pyMOL — **Genetic Annealing to Determine Protein Structures** — **github.com**
Monte Carlo Genetic Algorithm to determine protein structures through the optimization of Irbäck's off-lattice model energy equation (RMSD < 3.0).

Python — **Quadratic Unconstrained Binary Optimization** — **github.com**
Stochastic tunneling-enhanced simulated annealing for solving QUBO-type problems.

Python, NumPy — **aqua-blue** — **github.com/pypi.org**
Secondary maintainer of aqua-blue, a package building echo state networks.
- Implemented reservoir state time-lag, sparsity and spectral radius to improve prediction accuracy.
- Introduced timezone-aware datetime support for seamless interfacing with data pipelines.
- Demonstrated prediction capability by predicting quantum time evolution of Gaussian wave packets.

Python
NumPy
hyperopt

aqua-blue-hyperopt **github.com/pypi.org**

Primary maintainer of aqua-blue-hyperopt, a package implementing hyperparameter optimization functionality for aqua-blue models.

- Created modular hyperparameter optimization functionality for aqua-blue reservoir computers.
- Implemented custom loss function capabilities for flexible model evaluation.

Python
PyTorch
pytest

neurop **github.com/pypi.org**

Primary maintainer of neurop, a package implementing neural operators.

- Developed Fourier Neural Operator (FNO) architecture for solving PDEs.
- Implemented the Fractional Fourier Transform for use in Complex Neural Operators (CoNO).
- Built CI/CD pipeline with GitHub Actions for automated testing and deployment.

OPEN SOURCE CONTRIBUTIONS

QuTiP
Python

Quantum Toolbox in Python (QuTiP) **github.com**

Contributed to QuTiP, an open-source software for simulating quantum systems.

- Implemented functionality to return eigenkets as an operator in the Qobj class.

HONORS AND AWARDS

- Wielenga Research Scholarship, Honors College (2025-2026)
- Dean's List (2024, 2025)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)
- MSU President's Scholarship (2024)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)